

DATA EXPLORATION AND ANALYSIS

Project Objective: The Main objective of this project is to analyze and summarize dataset of the products by using basic excel functions, conditional and logical functions.

Dataset Description: The given dataset contains information about different kind of products and their Product ID, Brand Name, Quantity, categories and their prices.

Column Description:

These are the columns added to the given dataset.

Column Name	Price Range	Day	Country Code	Month
Description	Classifies product based on their price	Extract text using LEFT function	Extract text using RIGHT function	Extract text using MID function

DATA EXPLORATION:

1.Total Price of all products:289700

Formula used: =SUM(G2:G35)

2.Number of products:34

Formula used: =COUNTA(A2:A35)

3.Average Price:8520.588235

Formula used: Total price/number of products

4.Minimum price:150

Formula used:=MIN(G2:G35)

5.Maximum price:36000

Formula used:=MAX(G2:G35)

6.IF function:

A new column named price range is created.

Formula used:=IF(D2>=500) “High price”

If price is greater than or equal to 500 means it gives output as high price.

If price is less than 500 means it gives output as standard price.

7.SUMIF function:

To find the total price of products in electronics category, SUMIF function is used. It takes value of only electronics category and then sum all the values of that category.

Formula used:=SUMIF(F2:F35,F2,G2:G35)

8.COUNTIF function:

It counts the values whose prices are less than 100.

Formula Used:=COUNTIF(D2:D35,”<100”)

9.LEFT function:

A new column day has been created.

It takes two values from left in product Id column.

Formula used:LEFT(A2,2)

10.RIGHT function:

A new column country code has been created.

It takes two values from right in product Id column.

Formula used:RIGHT(A2,2)

11.MID function:

A new column month has been created.

It takes 4th to 6th values in product Id column.

Formula used:MID(A2,4,2)

Product I	Product Nam	Brand Nam	Price (\$	Quantity	Category	Total Price	Price Rang	Day	Country C	Month						
28-JAN-US	Laptop	Dell	1000	30	Electronics	30000	High Price	28	US	JA	Sum.Count Average					
15-FEB-US	Sneakers	Nike	80	15	Fashion	1200	Standard Price	15	US	FE	1.Total Price of all products:	289700				
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	5200	Standard Price	03	US	MA	A new column total price is created using formula (price*quantity)					
11-APR-US	Smartphone	Samsung	900	25	Electronics	22500	High Price	11	US	AP	To add the valu SUM(G2:G35) is used					
22-MAY-US	Backpack	North Face	70	20	Outdoor	1400	Standard Price	22	US	MA	2.Number of products:	34				
07-JUN-UK	Headphones	Sony	200	45	Electronics	9000	Standard Price	07	UK	JU	formula used: COUNTA(A2:A35)					
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	150	Standard Price	19	UK	JU	3.Average Price: 8520.588235					
23-AUG-UK	Blender	Ninja	90	35	Kitchen	3150	Standard Price	23	UK	AU	formula used: Total price/number of products					
05-SEP-UK	Tablet	Apple	500	50	Electronics	25000	High Price	05	UK	SE	4 Minimum Price: 150					
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	1300	Standard Price	14	UK	OC	Formula used MIN(G2:G35)					
17-JUN-IN	Laptop	HP	950	25	Electronics	23750	High Price	17	IN	JU	5 Maximum Price: 36000					
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	3600	Standard Price	25	AU	NO	Formula used: MAX(G2:G35)					
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	4200	Standard Price	08	DE	DE	6.IF Function:					
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	2250	Standard Price	18	CA	FE	A new column named Price Range is created					
16-APR-ES	Headphones	Bose	250	20	Electronics	5000	Standard Price	16	ES	AP	Formula Used: IF(D2>=500)"High Price"					
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	1750	Standard Price	21	CA	AU	IF(D2<500)"Standard Price"					
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	2400	Standard Price	20	CN	AU	7.SUMIF Function:					
27-JAN-IT	Laptop	Asus	980	10	Electronics	9800	High Price	27	IT	JA	Total Price of products in electronics category:	234200				
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	2250	Standard Price	01	UK	MA	Formula Used: SUMIF(F2:F35,F2,G2:G35)					
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	2000	Standard Price	14	US	AU	8 COUNTIF Function:					
14-MAY-RU	Camera	Nikon	700	50	Electronics	35000	High Price	14	RU	MA	Number of Products with price less than \$100	11				
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	1600	Standard Price	09	CA	JA	Formula Used: COUNTIF(D2:D35,"<100")					
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	4500	Standard Price	19	BR	JU	9 LEFT Function:					
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	1750	Standard Price	21	CA	AU	A new column named day has been created					
29-SEP-CA	Smartphone	Google	800	45	Electronics	36000	High Price	29	CA	SE	Formula Used: LEFT(A2,2)					
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	3250	Standard Price	03	CA	JU	10 RIGHT Function:					
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	16000	Standard Price	11	CA	JU	A new column named country code has been created					
16-APR-ES	Headphones	Bose	230	20	Electronics	4600	Standard Price	16	ES	AP	Formula Used RIGHT(A2,2)					
07-MAR-CA	Dress	Zara	60	30	Fashion	1800	Standard Price	07	CA	MA	11.MID Function:					
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	400	Standard Price	13	CA	AP	A new column named Month has been created					
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	650	Standard Price	24	CA	MA	Formula Used MID(A2,4,2)					
02-DEC-CA	Jeans	Levi's	50	50	Fashion	2500	Standard Price	02	CA	DE						
17-JUN-IN	Laptop	HP	950	25	Electronics	23750	High Price	17	IN	JU						
09-JUL-FR	Watch	Casio	100	20	Accessories	2000	Standard Price	09	FR	JU						

Summarization:

In this project basic calculations such as sum,average,count had been performed and minimum,maximum of the given dataset had been performed using various excel formulas.Logical and conditional functions for the dataset had been calculated.Text functions such as LEFT,RIGHT,MID had been used for data extraction.The given dataset had been analysed and summarized step by step above using various functions.

